SpamShield: One-Page Technical Summary

Team 1 · Pratap Gurung, Krushitha Janagama, Ganesh Babu Palacharla, Varsha Reddy Polla, Abhishek Virendra Pratap Singh · MSDS 630 Residency NLP Hackathon · University of the Cumberlands · October 2026

PROBLEM

A regional mobile carrier wants to block scam texts (fake deliveries, prize notices, bank impersonation) without ever blocking a real message from family or a doctor. False positives cost far more than missed spam, so we lead with spam precision and the false-positive rate, not accuracy: predicting “not spam” for everything already scores 87.6% accuracy.

DATA

UCI SMS Spam Collection: 5,574 English messages (13.4% spam). We removed 415 exact duplicates before splitting, leaving 5,159 messages (12.4% spam), split stratified into 3,611 train, 774 dev and 774 test (seed 42). The TF-IDF vectorizer was fit on train only, the decision threshold was tuned on dev, and test was scored once.

METHOD

1. **Baseline:** Naive Bayes written from scratch (lowercase, links as one token, add-1 smoothing, log-space scores).
2. **Comparison:** scikit-learn Naive Bayes and logistic regression on word and bigram TF-IDF.
3. **Tuning:** threshold chosen on dev for a false-positive rate of at most 0.5%, then applied once to test.
4. **Robustness:** leetspeak spellings with and without a digit-to-letter normalizer; 7 hand-written modern scam-style messages.

RESULTS (TEST SET: 96 SPAM, 678 REAL MESSAGES; SPAM IS THE POSITIVE CLASS)

Model	Precision	Recall	F1	False-positive rate
Naive Bayes (scratch), threshold 0.5	0.944	0.885	0.914	0.74%
Naive Bayes (scratch), dev-tuned threshold	0.924	0.885	0.904	1.03%
scikit-learn Naive Bayes, TF-IDF	1.000	0.698	0.822	0.00%
Logistic regression, threshold 0.5	1.000	0.875	0.933	0.00%
Logistic regression, dev-tuned threshold	**0.989**	**0.938**	**0.963**	**0.15%**

Shaded row = best F1. With 5 versus 1 false positives out of 678, the 95% intervals overlap (0.32–1.71% vs 0.03–0.83%), so the ranking is suggestive, not proven. Leetspeak cut Naive Bayes spam recall from 0.885 to 0.833; the normalizer restored 0.885 with no extra false positives.

LIMITATIONS

- **Dataset age.** The data is older, largely UK and Singapore text. On 7 modern scam-style messages we wrote, the model missed a toll-payment scam and blocked an appointment confirmation (anecdotal).
- **Threshold transfer.** The dev-tuned Naive Bayes threshold missed its target on test (0.44% false positives on dev, 1.03% on test); 92.8% of its probabilities are above 0.99 or below 0.01.
- **Evaluation noise.** Exact-duplicate removal misses near-duplicates, and a single test split is noisy: one extra false positive moves the rate by 0.15 points.

NEXT STEPS

Collect 30 to 50 modern labeled scam messages; calibrate probabilities and choose the threshold by cross-validation; add features for phone numbers and per-minute prices; fine-tune DistilBERT as the stretch goal.